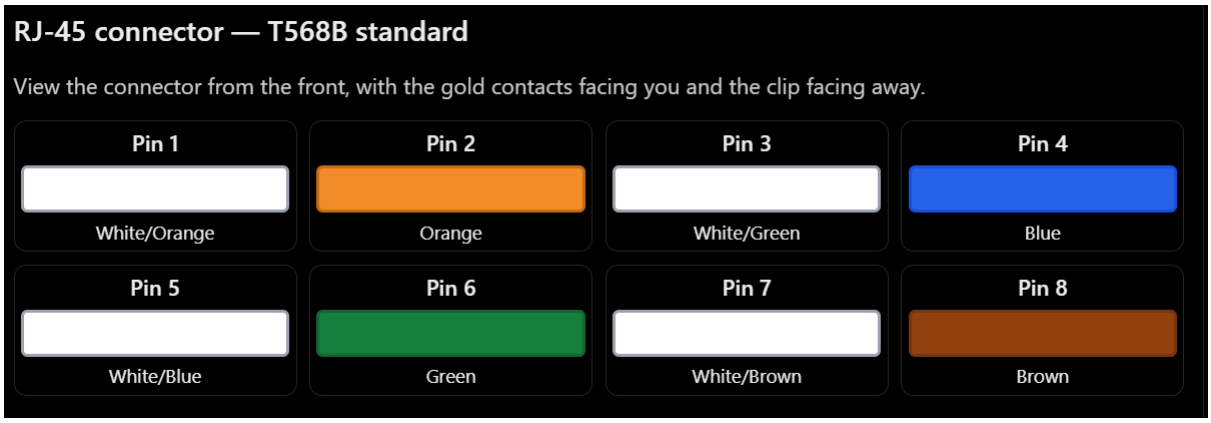
Network Cables and Connectors

1. Cat6 Ethernet Cable: T568B Wiring Diagram



Correct T568B color order

1. White/Orange
2. Orange
3. White/Green
4. Blue
5. White/Blue
6. Green
7. White/Brown
8. Brown

The RJ-45 connector plugs into an Ethernet port on devices such as routers, switches, computers, and gaming consoles.

2. Three Types of Network Cables

Cable Type	**Example Device or Service**	**Why It Is Used**
Cat5e Ethernet	Wi-Fi router connected to a computer	Supports speeds up to 1 Gbps over distances up to 100 metres under standard conditions.
Cat6 Ethernet	Gaming PC or Zomato restaurant POS terminal	Provides reliable wired connectivity and supports up to 10 Gbps over shorter distances, typically up to 55 metres under suitable conditions.
Fiber-optic cable	IPL streaming server or ISP backbone	Carries data as light, supports high bandwidth, and works well over long distances.

3. Steps to Crimp an RJ-45 Connector

Watch this YouTube tutorial: How to Crimp Cat5/Cat6 Network Patch Cables. It demonstrates cable preparation, T568B wiring, crimping, and testing.

Video Highlight+1

Tools required: Cat6 cable, RJ-45 connector, wire stripper, crimping tool, and LAN cable tester.

Steps:

1. Cut the Ethernet cable to the required length.
2. Carefully remove about 2–3 cm of the outer jacket without damaging the inner wires.
3. Separate and straighten the eight coloured wires.
4. Arrange the wires in T568B order: White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.
5. Trim the wires evenly so they fit inside the connector.
6. Insert all eight wires into the RJ-45 connector, ensuring they reach the front contacts and the cable jacket extends inside the connector.
7. Use the crimping tool to secure the connector.
8. Repeat the process at the other end using the same wiring standard.
9. Test the cable with a LAN cable tester to check that all eight connections are correct.

Two common mistakes to avoid:

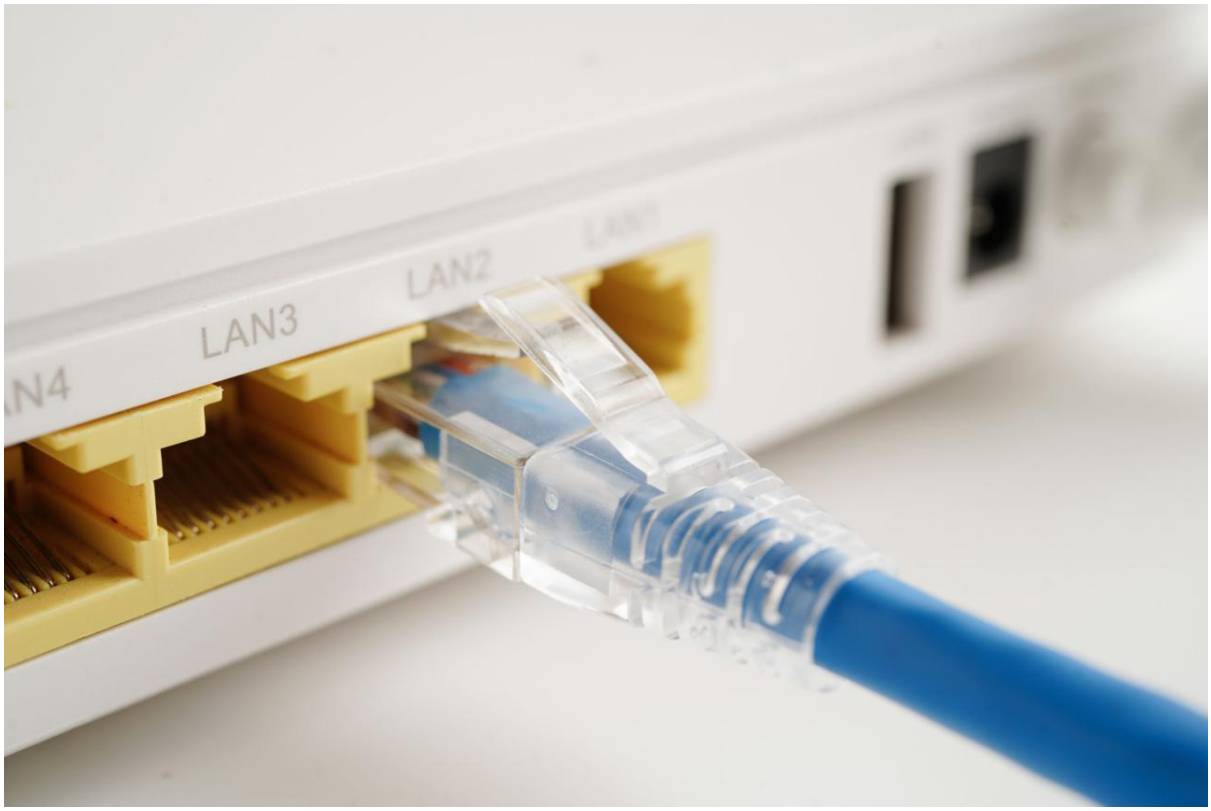
- Incorrect wire colour order, which can cause the cable to fail.
- Wires not inserted fully into the connector, resulting in poor electrical contact.

4. Best Cables for an IPL Screening and Gaming Zone

Connection	**Recommended Cable**	**Connector**
Router to network switch	Cat6 Ethernet	RJ-45
Switch to gaming consoles or PCs	Cat6 Ethernet	RJ-45
Switch to nearby smart TVs	Cat6 Ethernet	RJ-45
Main network link between distant areas	Fiber-optic cable	LC or SC, depending on equipment

Reason: Cat6 Ethernet provides high-speed, low-latency, reliable connections for gaming consoles and streaming devices. Fiber is suitable for long-distance links or high-bandwidth backbone connections. Using a network switch allows multiple screens and consoles to connect to the same LAN.

5. Photo of a Network Cable in Use



Example photo of an Ethernet cable connected to a router.

Description: The photo shows a blue Ethernet cable with a transparent RJ-45 connector plugged into a router's Ethernet port. The cable carries data between the router and a connected device, such as a computer or network switch, providing a wired network connection.